Varuna FloodTwin: an honestly-validated, free-tier flood digital twin with street-routed drainage planning for Indian cities

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Abstract

Urban waterlogging routinely paralyses Indian cities, yet municipal drainage planning rarely has access to calibrated hydraulic models. We present *Varuna FloodTwin*, an end-to-end open pipeline that builds a differentiable urban flood twin for any area of interest from free satellite data (FABDEM, ESA WorldCover, JRC surface water, Sentinel-1), trains a millisecond what-if emulator, and serves both through a public web dashboard — with every stage running on free tiers (Colab/Kaggle GPUs to build, CPU to serve). The system covers three cities — Patna, Bengaluru, and Mumbai (BMC, as four exactly-tessellating 256^2 tiles with a city-wide view) — and adds a live participatory layer: citizen flood reports that feed a reward-gated nightly fine-tuning loop, live-forecast alerts, exact storm water-balance accounting, grounded multilingual advisories, and a satellite power-outage overlay. We report five findings. **(1) A flood CSI is uninterpretable without its trivial baselines:** on 18 Sentinel-1 storms across two cities, a parameter-free climatology — "the cells that were wet in most other storms" — scores 0.30–0.58, above any physics-model CSI we are aware of in this literature, while all-wet scores 0.013–0.047. Against that scale our dynamic twin's 0.054 (re-measured; we previously reported 0.041 on a test set that included a scene with no ground truth) is a weak but real signal, and gradient calibration of per-land-class roughness/infiltration remains near-null (held-out CSI $0.048 \to 0.049$): the model predicts a comparable flooded *area* but not its *location*. **(2) That failure is specific to routing, not to the terrain data — correcting our own earlier claim:** a U-Net reading the same DEM, land cover and soil rasters reaches CSI 0.290 ± 0.006, $5.4\times$ the twin on identical scenes, because relative micro-relief survives the spatially-correlated DEM error that destroys absolute elevation. Under a ± 1 m DEM ensemble total flooded area is stable (Patna $10.79 \pm 0.35\,\text{km}^2$) but only 5% of it is co-located across members on deltaic flats (52% in hillier Bengaluru), which is why a downhill solver cannot place water — yet the learned model still ties the parameter-free climatology (0.288), so what terrain buys on a *known* city is nothing, and its real value is transfer: 0.094 ± 0.021 on a city never seen in training, where a climatology cannot be computed at all. **(3) Planning can survive location uncertainty:** intervention value is *measured by re-simulation* rather than assumed. A dense storm-drain "spiderweb" routed along the real OpenStreetMap street graph — $\sim$40 inlets, one reverse multi-source Dijkstra from safe low-ground outfalls (never the river, which runs high during floods) — cuts simulated street flooding by 73–89% in the Patna and Bengaluru windows and 33–63% across the four denser Mumbai tiles at 100 mm/day, versus 21–26% for sparse least-cost canals, while *improving* cost-efficiency (INR 850 vs INR 1,309 per m^3 removed in Patna). A complementary distributed-storage planner sites detention containers only where they can physically go (no building footprints, no permanent water, under-street allowed) and emits phased, indicatively-costed build-out plans; its top-ranked sites land on each city's notorious flood spots (Koramangala in Bengaluru; the Mithi at Kalina in Mumbai) with no city-specific tuning. **(4) Street-level flood knowledge is learnable and city-transferable:** a graph neural network over the OSM street graph, trained only on labels the twin generates itself and FiLM-conditioned on rainfall, amortizes the raster pipeline into a millisecond per-street risk ranking at any storm size (edge AUC 0.89–0.94 on held-out storms), yields evacuation routes that recover 65% of an oracle's floodwater avoidance at 57% of its detour, and — its features being local and per-graph standardized — transfers zero-shot between the flat and the hilly city (AUC 0.72–0.85), the learned counterpart of finding (2). **(5) Determinism can lock in failure:** because our dataset generator reseeds the global RNG with a fixed seed, the emulator received a bit-identical initialization on every rebuild — and on the half-ocean Mumbai Island-City tile that exact initialization *deterministically* collapses to "never flood" (the optimizer drives the softplus head into its zero-gradient regime; the tell is a validation RMSE frozen to the last digit across epochs). Naive retries can never fix a seeded collapse; a detect-and-reseed guard turns the worst tile into the best-fitting emulator in the system (5.0 cm validation RMSE). Code, artifacts, and the live dashboard are public.

1 Introduction

Pluvial (rain-driven) waterlogging is among the most frequent climate impacts in South Asian cities, and it is worsening under climate change. Yet the planning reality in most Indian municipalities is a drainage map that predates the city's growth, no calibrated hydraulic model, and no budget for commercial modelling. Tools that could plausibly change decisions must therefore be (i) buildable from *free, global* data for *any* area of interest, (ii) cheap enough to run interactively, and (iii) honest about what they can and cannot claim.

This paper describes such a system and, we argue, an appropriately honest way to frame its skill. Our contributions:

1. **A reproducible satellite→twin→dashboard pipeline** on free infrastructure: a differentiable 2-D inertial flood twin [1] built per-city from FABDEM [2], ESA WorldCover [3] and JRC surface water [4]; a U-Net-style emulator [5] for millisecond what-ifs; a public FastAPI + React-Leaflet dashboard with live rainfall sliders, intervention planners, exposure views and a grounded LLM brief. Deployed for three cities (nine bundles): Patna and two sub-windows, Bengaluru, and Mumbai (BMC) as four exactly tessellating 256^2 tiles with a city-wide aggregate view.
2. **An honest validation** against Sentinel-1 SAR [6] that places the twin *and* classical terrain indices (TWI [7], HAND [8]) in the same low-CSI band on deltaic terrain, plus a DEM-perturbation ensemble that explains *why* — a diagnosis we believe transfers to other flat megacity floodplains.
3. **Street-graph drainage planning:** storm drains routed on the real OSM road network [9] with gravity-aware costs and flood-safe outfall selection (low-lying non-built ground rather than the river), whose benefit is measured by re-simulating the carved terrain. Density is the decisive lever: a ∼40-inlet spiderweb removes 4× more flood volume than 2–5 sparse canals at *lower* cost per m^3.
4. **Self-supervised street-graph learning (FloodGNN):** an edge-conditioned message-passing network [10] on the same street graph, FiLM-conditioned on rainfall [11], trained purely on labels the twin generates (emulator depth sweeps, drain-planner flows). One forward pass ranks every street's flood risk at any rainfall and powers a public flood-safe routing endpoint; a hold-one-city-out experiment extends finding 2 to learned models — street-level flood knowledge transfers across cities even where cell-level locations do not.
5. **Buildability-constrained, phased intervention portfolios:** detention containers sited only on physically available ground (building footprints and permanent water excluded via rasterized OSM and WorldCover; under-street detention allowed, as Indian cities actually build it), each site sized to its own local pooled depth, with the flood cut *re-measured* after placement and delivered as a 10/20/40%-cut phase ladder with indicative civil-works costs. Buildability costs some hydraulic efficiency on flats and we report that honestly; in dense terrain it *improves* drain hydraulics (streets trace drainage valleys).
6. **A live participatory layer with an auditable learning gate:** rate-limited citizen flood reports [12] persisted to an open dataset; a nightly loop that converts reports plus observed rainfall into supervision (day-wise splits, dry-day filtering, per-cell medians) and fine-tunes the emulator behind a *reward gate* (band-tolerant depth loss, simulation-replay anchor, bootstrap win-rate) so a model update ships only if it is measurably better and provably not worse; live-forecast alerts, exact storm water-balance accounting (closure to $\sim 10^{-6}$ relative), grounded English/Hindi/Marathi advisories, and a quality-masked VIIRS Black Marble [13] power-outage overlay.

2 System

Build layer (GPU, run once per city). For an area of interest, Google Earth Engine exports FABDEM (30 m, forests/buildings removed; Copernicus GLO-30 fallback), ESA WorldCover 10 m classes, JRC permanent-water occurrence, and Sentinel-1 GRD scenes. The twin domain is an $N \times N$ grid at 60 m — 128^2 (a 7.7 km window) for the Patna-family and Bengaluru bundles, 256^2 (15.4 km) for each Mumbai tile; the four tiles share exact lat/lon edges, so city-wide sums never double-count (tiles are independent closed-boundary models: cross-seam flow and tides are not simulated, and the dashboard says so). Terrain analysis (depression filling, sink catchments) yields candidate detention sites; WorldCover classes map to Manning roughness and infiltration tables. The simulator is the inertial shallow-water scheme of Bates et al. [1] implemented in PyTorch, so every simulation is differentiable end-to-end; rainfall forcing uses Open-Meteo reanalysis/forecast [14]. A flood-weighted U-Net emulator distils the twin for millisecond dashboard queries (wet cells weighted $\times 20$ in the loss — plain MSE collapses to "never flood" because flooding is sparse; validation RMSE 5–21 cm across the nine bundles, monotone dose-response). The same instrumented solver yields an *exact* storm water balance — with closed boundaries, rain = infiltration + ponding to $\sim 10^{-6}$ relative error — which powers a public "where does the rain go?" panel including an all-impervious counterfactual (the ponding avoided by existing soil and vegetation).

A reproducibility hazard, and its guard. Our dataset generator seeds the global RNG for reproducible storm sampling — which silently makes the *network initialization* bit-identical on every rebuild. On the Mumbai Island-City tile (19% of cells flood-prone; much of the window is harbour), that particular initialization deterministically collapses: within one epoch the optimizer pushes the softplus output head so far negative that its gradient vanishes, and the emulator is permanently stuck predicting 0 m everywhere while the whole-grid validation RMSE looks plausible (16.3 cm — exactly the RMS of the targets) and stays *frozen to the last digit* across 40 epochs: the signature of exactly-zero gradients. Because the failure is seeded, rebuilding "for luck" can never fix it — we measured three bit-identical collapses before diagnosing it. The shipped trainer now detects a collapsed validation prediction (no cell above the wetness threshold despite wet targets) and retrains from explicit alternate seeds; the first alternate seed took this tile from dead to the best-fitting emulator of the nine (5.0 cm validation RMSE). We flag the pattern — *fixed seeds make failures reproducible too* — because build pipelines that reseed globally for data reproducibility are common, and a frozen validation metric is an easy, general tell.

Serve layer (CPU, free hosting). A FastAPI backend (Hugging Face Space, Docker) serves the committed per-city artifact bundles — including cached OSM road graphs and exposure, so the deployed service needs no external API access — and re-runs interventions live in seconds. The React-Leaflet frontend (Vercel) exposes rainfall what-ifs, ward alerts, drainage/storage/excavation planners, building- and road-level exposure, validation panels, and an LLM planning brief grounded in the bundle's numbers. Version 2 adds the live layer of §6: citizen reports, live-forecast mode, the water-balance panel with a storage-container designer, multilingual advisories, night lights, and the nightly learning log. Everything (build to serve) runs within free tiers; the total build for a new city window is ∼15–25 minutes on a free T4, including all serve artifacts.

3 Honest validation: what a CSI is worth here, and what terrain can actually predict

An earlier version of this section reported that the dynamic twin scores CSI 0.041 against Sentinel-1, that static topographic indices score the same, and concluded that *no* topographic router can co-locate ponding in this data regime. Two of those three statements survive re-measurement. The third does not, and we correct it here, because the experiment that refutes it is our own.

Protocol. Sentinel-1 water masks for 18 monsoon storms across two domains (Patna, 2023–2025, 7 storms; Mumbai-Northeast, 2025–2026, 11 storms) are scored on a common 60 m grid with permanent water (JRC $\geq$50%) masked *on both sides*. Every method — physical, index-based and learned — sees the identical grid, mask and max-pooled truth, and every method's detection threshold is chosen on the *other* storms of the fold, never on the scene being reported. One Patna date (2024-07-07) has zero observed wet cells, hence no ground truth, and contributed a forced 0.0 to every method's mean in our earlier table; it is dropped. The scoring path is verified by re-deriving that earlier table from it under the earlier protocol (twin 0.0412 $\rightarrow$ 0.0409, TWI 0.0422 $\rightarrow$ 0.0429, HAND-lite 0.0505 $\rightarrow$ 0.0504).

Table 1: What CSI is worth on Sentinel-1 flood extent, per domain, with no model. "Climatology" predicts the cells that were wet in most *other* storms of that domain: no rainfall, no physics, no parameters. "SAR vs SAR" scores every date against every other date of the same domain — an empirical agreement ceiling between two radar passes of the same city.

domain	storms	all-wet	random	climatology	SAR vs SAR
Patna	8	0.047	0.025	**0.300**	0.160
Mumbai-Harbour	11	0.021	0.010	**0.581**	0.462
Mumbai-Northeast	11	0.013	0.007	**0.391**	0.271

Table 2: All methods on the identical 18 storms, thresholds chosen off the reported scene. The learned model is a 7.9 M-parameter U-Net over 23 terrain channels with rainfall ablated, 3 seeds. Every terrain method is quoted at its most favourable threshold rule.

method	mean CSI	bias	POD
random	0.014	1.00	—
static depression depth	0.032	—	0.21
all-wet	0.029	—	1.00
TWI (topographic wetness)	0.039	—	0.26
HAND-lite (height above drainage)	0.039	—	0.90
dynamic twin (antecedent rain)	0.054	—	0.29
learned (terrain only)	$\mathbf{0.290 \pm 0.006}$	1.39	0.51
climatology (no model, no rain)	0.288	1.29	—

3.1 A CSI without its trivial baselines is uninterpretable

The single most useful thing we can report is not a model score. It is the range a flood CSI can occupy on this task *before any model exists* (Table 1).

A parameter-free climatology scores 0.30–0.58. That is higher than any CSI we are aware of being reported for a physics-based urban flood model on Sentinel-1, including ours. The reason is that the Sentinel-1 target is dominated by storm-independent water the JRC permanent-water mask does not remove — tidal flat, mangrove (WorldCover class 95 is present in both Mumbai tiles), seasonal pond, paddy. CSI on this task has therefore mostly been measuring *can you find standing water*, which the twin never attempted: it predicts only the storm increment.

The signature confirms the reading. On Mumbai-Northeast's one true flood (2026-07-08, 2 937 wet cells against a ∼450 baseline) the climatology scores its *worst*, 0.075. It wins the quiet dates and loses the flood. We recommend that CSI in this literature always be reported beside all-wet and a leave-one-out climatology; the numbers below are meaningless without them, and so, in retrospect, was our 0.041.

3.2 The twin, re-scored, and a correction in our own disfavour

Table 2 carries three results.

The twin is better than we published. On a fair test set it reaches 0.054, not the 0.041 we reported — dropping the dead scene and adding a second city both help it. It also clears all-wet (0.029) and every static index, which our Patna-only table did not show. We state this because the correction runs against our own headline.

Co-location *is* learnable from terrain, so our "terrain-fundamental" claim was wrong. The same DEM, the same land cover, the same soil rasters, read by a U-Net instead of routed by a solver, reach 0.290 ± 0.006 — 5.4× the twin and 9.9× all-wet, on identical scenes. What fails in this regime is *routing* water over a noisy DEM, not *predicting* where water goes. The feature that makes the difference is available to both: elevation minus its own Gaussian blur at 2/8/32 cells. DEM error is spatially correlated, so relative micro-relief survives a bias that destroys absolute elevation — the same ±1 m error the ensemble below injects.

But the learned model ties the climatology it was meant to beat. 0.290 against 0.288, a gap of 0.0014 against a seed standard deviation of 0.0064. On a city with a Sentinel-1 archive, a neural network over 23 terrain channels is worth nothing over "these cells are usually wet". The learned model's contribution is therefore *not* accuracy; it is transfer, which we isolate next.

3.3 The deployable claim is transfer, not accuracy

Held-out *city* (leave-one-domain-out, 3 seeds, 3 domains), the model never having seen the target terrain: CSI 0.094 ± 0.021 at bias 1.07. That is 3.6× all-wet and roughly 2× the twin, on cities it has never seen. The climatology scores 0.396 there and wins — but it does so using the held-out city's *own* Sentinel-1 history. For a city with no radar archive the climatology cannot be computed at all, and the learned model is the only method in Table 2 that runs. This is the claim that matters operationally, and it rests on three cities, which is thin; we report it as a floor rather than an estimate.

3.4 Three levers that are not levers

Capacity is not the bottleneck. Held-out floods, terrain only: CSI 0.221 at 2.0 M parameters, 0.232 at 7.9 M, 0.221 at 31.4 M, 0.219 at 70.6 M. Flat across a 35× range, entirely inside seed noise.

Resolution is not the bottleneck. Our standing explanation for poor co-location was that real ponds are $\sim$810 m^2 (28.5 m across) and were being scored on 60 m cells. Re-running at 30 m quadruples the cell count and changes nothing: the skill multiple over all-wet is 10.4× at 30 m against 9.7× at 60 m. (Raw CSI is not comparable across grids — the base rates differ — which is why the multiple is the column to read.)

Rainfall is not usable, and it actively hurts. Ablating the rainfall channels *improves* the model's ranking quality by 0.124 at $\approx$7 seed standard deviations, with three-seed ranges disjoint. A permutation control that preserves every rainfall distribution and destroys only the storm-to-rain pairing lands monotonically between real rain and no rain ($0.184 \rightarrow 0.272 \rightarrow 0.313$): real rainfall is strictly the worst of the three inputs, because the network fits a rain-to-wetness mapping on the training storms that does not transfer. Two alternative explanations were tested and refuted — tidal contradiction between the two Mumbai tiles (predicted that dropping the tidal tile would stop the harm; it did not), and failure to extrapolate to larger storms (rain costs 0.09 CSI in-distribution too). The effect has one boundary: on an unseen city, rainfall neither helps nor hurts (0.0939 vs 0.0936), so the claim is that rainfall lets the model shortcut a domain it has already seen, not that rainfall is universally harmful.

Two properties of the rainfall archive explain why there is nothing to extract. Nine probe points spanning a 15 km tile collapse to a single archive cell, so spatial rainfall is not merely unused but unavailable. And the source is not what we previously stated: Open-Meteo's archive endpoint, queried without an explicit `models` parameter, returns ECMWF IFS at 9 km, not ERA5 — 554.2 mm against ERA5's 1,213.4 mm over the same window and point, a factor of 2.19 that exceeds any effect our physics calibration produced. The twin was never calibrated on ERA5. We now pin the model explicitly, which reproduces every number in this paper exactly.

3.5 Why the physics router fails: a DEM-uncertainty diagnosis

We re-simulate under a 10-member ensemble of spatially-correlated DEM perturbations ($\sigma = 1$ m, 5-cell correlation length — FABDEM's nominal error class). Total flooded area is robust: Patna 10.79 ± 0.35 km^2 at 100 mm. But only 0.58 km^2 (5%) floods in $\geq$90% of members. In Bengaluru, with 10× steeper relief, 4.56 of 8.82 km^2 (52%) is robust (Figure 1). Gradient calibration of 16 per-WorldCover-class Manning/infiltration multipliers moves held-out CSI only $0.0483 \rightarrow 0.0488$, with multipliers staying in $[0.94, 1.14]$: roughness and infiltration change *how deep*, not *where*. We also ruled out a georeferencing artifact — mirroring the SAR mask doubles CSI, but the DEM is provably correctly oriented (corr($-$DEM, JRC) = 0.80 unflipped vs 0.05 flipped; the Ganga sits 7.8 m below domain mean), so the flip is a coincidental mirror and is not applied.

The scope of this diagnosis is narrower than we previously claimed. On a floodplain where metre-scale DEM noise exceeds the relief that decides which street ponds, a solver that *routes* water downhill cannot place it reliably, and its calibration gradient with respect to co-location is ≈ 0. That is why the twin sits at 0.054. It is *not* a proof that the terrain data lacks the information: a model reading relative micro-relief from

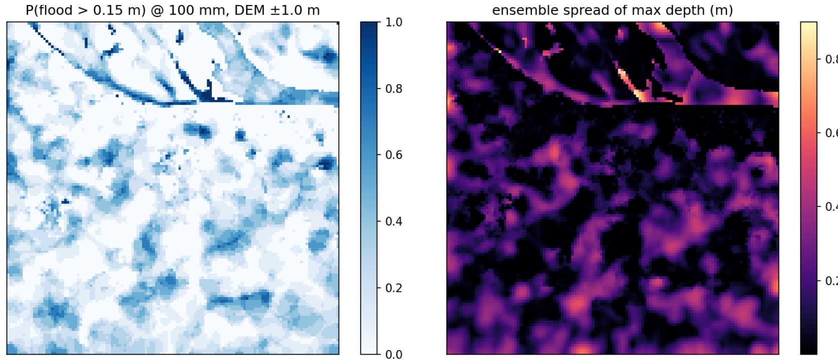

Figure 1: DEM ± 1 m ensemble: flood probability over 10 members (Patna, 100 mm). Total area is stable but its location shifts — only 5% of flooded area is common to $\geq$90% of members on these deltaic flats (52% in hillier Bengaluru).

the same rasters reaches 0.290. The correct statement is that absolute elevation is too noisy to route over, while relative elevation is not too noisy to learn from. It follows that (a) reported skill in this regime should always be accompanied both by such an ensemble and by the trivial baselines of Table 1, and (b) planning tools should be judged on decisions that are robust to it, which we turn to next.

4 Planning that survives location uncertainty

All interventions below are evaluated the same way: modify the terrain, re-simulate the design storm (100 mm/24 h), and measure flooding on built-up cells, *excluding* the intervention's own cells from both the before and after masks (relocating water onto a drain does not count as reduction; the excluded fraction is reported and stays below a few percent of built cells).

Street-graph storm-drain spiderweb. Sparse "worst-basin to nearest outfall" canals are how our v2 planner (and, anecdotally, much real retrofitting) works: 2–5 least-cost corridors over the DEM grid cut 20.8% (Patna) / 25.7% (Bengaluru), where preferring OSM road cells and avoiding buildings *improved* the Bengaluru cut 3× — streets trace drainage valleys in dense terrain, so buildability and hydraulics align. Scaling this idea up, we route a dense network on the actual street graph: the domain's OSM roads (fetched once via Overpass, cached per bundle; 42.8k nodes / 6.8k ways for Patna) form a graph whose nodes carry DEM elevations; edge cost is street length plus an uphill penalty of 2,400 m per metre of climb (the grid router's $dx(1 + 40\,\Delta z)$ expressed per metre). One reverse multi-source Dijkstra from all outfalls labels every street with its gravity-cheapest exit; ∼40 inlets placed at the deepest cells of each flood basin (greedy with suppression, basin volume apportioned by local depth) then trace their exit chains, whose union is a forest with shared, flow-accumulating trunks. Outfalls embody a flood-safety principle: *never the river* — during a flood it runs high — but the lowest *safe* ground (non-built, buffered from buildings and from the river corridor), downhill domain exits, and detention pits, with pits handicapped by 2 km of equivalent cost so that conveying water away wins wherever gravity allows. The chosen streets are carved into the DEM as 2 m channels with strictly descending beds (shared trunks take the deepest bed, provably still monotone), Manning $n = 0.02$, and the storm is re-simulated.

Results. Across the eight areas the spiderweb cuts street flooding by 80.9% (Patna), 89.4% (Patna-East), 80.3% (Patna-West), 72.6% (Bengaluru), and 63.0/50.2/39.2/32.5% on the Mumbai South/East/West/North tiles, versus 21–26% for the sparse plans (Table 3, Table 4, Figure 2); dose-response over 25–200 mm is monotone. Density improves *economics* as well as efficacy: INR 850/m^3 removed versus INR 1,309/m^3 for the sparse plan (Patna; Bengaluru INR 1,121/m^3), because shared trunks amortise length across basins. The lower Mumbai cuts are informative rather than disappointing: the tiles are 4× larger, far denser, and partly harbour, so a fixed ∼40-inlet budget covers a smaller fraction of basins — cut scales with inlet density, which is exactly the lever a municipality controls. The exposure layer grounds the maps in assets: at 100 mm, 1,205 of 3,749 OSM buildings and 3,036 of 6,762 road segments in the Patna window are at risk; in Mumbai, the Western-suburbs tile is the worst-exposed with 15,449 of 31,597 buildings at risk — consistent with the city's

Table 3: Measured intervention ladder, Patna @100 mm (re-simulated; costs at indicative Indian civil-works rates: INR 9,000/m drain, INR 300/m^3 excavation, INR 6,000/m^3 RCC storage). The dense street-routed network dominates sparse canals on *both* axes.

intervention	measured cut	flood removed	cost per m^3 removed
gradient-optimised excavation (8 sites)	4.1%	0.06 M m^3	INR 758
detention pits only (8 sites)	7.8%	0.11 M m^3	—
sparse canals + pits (v2 grid, 3.2 km)	20.2–20.8%	0.28 M m^3	INR 1,309
street spiderweb + pits (40 inlets, 44.5 km)	**80.9%**	**0.87 M m^3**	**INR 850**
distributed storage, 727 sited micro-basins	30%	0.43 M m^3	∼INR 9,200

Table 4: Three-city summary at the 100 mm design storm. Spiderweb cut: measured street-flood reduction of the street-routed drain network. Buildable sites: ranked container list emitted (share under streets). Phase 1: sites and indicative cost to a 10% measured cut (INR 6,000/m^3 RCC detention). Dense tiles reach 10% only — honest ladders, not aspirational ones.

area	grid	spiderweb cut	buildable sites (street)	phase-1: 10% cut
Patna	128^2	80.9%	1,500 (87%)	193 sites, INR 165 cr
Patna-East	128^2	89.4%	410 (80%)	50 sites, INR 74 cr
Patna-West	128^2	80.3%	1,500 (78%)	150 sites, INR 127 cr
Bengaluru	128^2	72.6%	273 (84%)	176 sites, INR 242 cr
Mumbai South	256^2	63.0%	1,071 (63%)	231 sites, INR 223 cr
Mumbai West	256^2	39.2%	1,500 (68%)	493 sites, INR 825 cr
Mumbai East	256^2	50.2%	1,463 (80%)	365 sites, INR 403 cr
Mumbai North	256^2	32.5%	1,056 (77%)	139 sites, INR 373 cr

lived flood geography (Andheri–Kurla, the lower Mithi).

Buildable, phased distributed storage. The adaptive storage planner answers "how many detention containers, and *where*, to cut the flood by X%?" Candidate cells are the dynamic local minima of the simulated flood (ranked by pooled depth), each site is sized to the water it actually holds, and the cut is re-measured by simulation. Version 2 makes the sites *buildable*: building footprints (rasterized OSM) and permanent water are excluded from candidacy — you cannot dig a tank under an occupied house or in the harbour — while under-street detention remains allowed, mirroring how Indian cities actually retrofit (e.g. BMC's under-road Hindmata tanks). Buildability is not free on flats: excluding building-footprint minima removes some of the deepest candidates, and the reachable cut ladder drops accordingly (we report it, rather than quietly siting tanks inside houses). Each area ships its explicit ranked site list (up to 1,500 sites with coordinates, per-site volume, and an under-street flag; 63–87% of sites land under streets) and a 10/20/40%-cut phase ladder costed at indicative RCC-detention rates (Table 4). The planner has never seen a flood report or a news archive, yet its top-ranked sites fall on each city's chronically flooded places — Koramangala in Bengaluru, the Mithi at Kalina in Mumbai-West — a qualitative check that deep simulated minima on buildable ground coincide with where flooding actually hurts (Figure 3).

Why we believe these decisions despite §3. The quantities that drive the plans are the ones the ensemble shows to be robust: total flooded volume, basin existence and approximate depth ordering, and downhill topology — not the exact street that ponds first. The planner's outputs are also *portfolios* (many inlets, many micro-basins) whose value degrades gracefully if any single pond shifts. We therefore frame outputs as relative planning guidance (which strategy, what density, where first), not as certified absolute depths.

5 Learned street-level risk, routing, and cross-city transfer

Both planning surfaces above route on graphs with hand-set costs: storm drains via a reverse Dijkstra whose flow-direction cost ($\ell + 2400\,\Delta z^{+}$ m) encodes a fixed uphill aversion, and evacuation implicitly via raster overlays recomputed per storm. We replace the hand-set cost with a learned one. **FloodGNN** is an

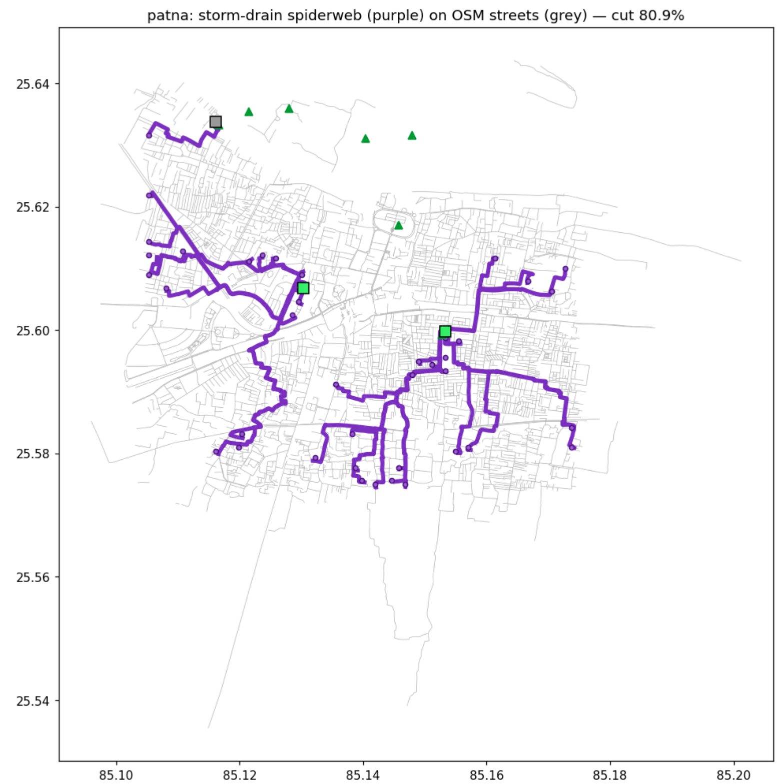

Figure 2: The planned storm-drain spiderweb for Patna (purple; line width $\propto$ accumulated flow) traces real OSM streets (grey) from 40 flood-basin inlets to detention pits and safe low-ground outfalls. Measured cut: 80.9% of street flooding at 100 mm.

edge-conditioned message-passing network [10] (4 rounds, 96-dim states, ~439k parameters, plain PyTorch) over the OSM street graph (37,017 nodes / 79,928 directed edges for Patna), FiLM-conditioned on rainfall [11] so a single model covers the continuous storm range. Node features are deliberately local and standardized per graph — elevation anomaly, pit-ness, grade, flow accumulation, permanent water, SAR waterlogging frequency, building/road density — so nothing identifies the city. Supervision is generated by the twin itself: the emulator's depth fields for a 20–240 mm sweep sampled at street nodes and edges, and the drain planner's per-street conveyed volume at three (rain, inlet-count) configurations (~9 s of label generation per city, CPU). Wet targets carry the same ×20 loss weight as the emulator (flooding is sparse; §2), validation holds out *storm sizes* rather than nodes, and the shipped checkpoint is the best-validation epoch.

FloodGNN answers three questions. *Risk*: per-street flood depth at any rainfall in one forward pass — edge AUC 0.906 on held-out storm sizes in Patna and 0.89–0.94 across the four areas, vs 0.856 for a 24-epoch CPU smoke model trained on Patna alone. *Routing*: cost-inflated Dijkstra ($\ell \cdot (1 + 8\min(d, 1.5) + 25 \cdot [d > 0.3\,\mathrm{m}])$) yields evacuation routes crossing 1,570 m of flooded street per route versus 2,297 m for the flood-ignorant shortest path and 1,172 m for an oracle reading the emulator directly, at a 21% mean detour against the oracle's 37% (Patna at 140 mm, 150 random origin–destination pairs, all judged on oracle depths): 65% of the oracle's improvement at 57% of its detour. The pattern holds in every area — in Bengaluru the GNN route even crosses *less* water than the oracle router's (1,172 vs 1,333 m), since the oracle minimises risk-inflated cost, not wet metres (Figure 4). *Placement*: the flow head recovers the drain planner's trunk ranking only weakly (Spearman $\rho = 0.14$–0.24) — usable for corridor suggestion in the dashboard, but trunk placement remains the planner's job, and we report the number rather than the claim.

Cross-city transfer. Trained with Bengaluru held out entirely — an all-flat training set — FloodGNN ranks flooded Bengaluru streets at AUC 0.721 zero-shot, against 0.916 when Bengaluru is trained on. The reverse direction is stronger: held-out Patna scores 0.847 zero-shot versus 0.906, because the mixed flat-plus-hilly training set covers Patna's regime. A no-message-passing ablation (layers = 0: the same features through

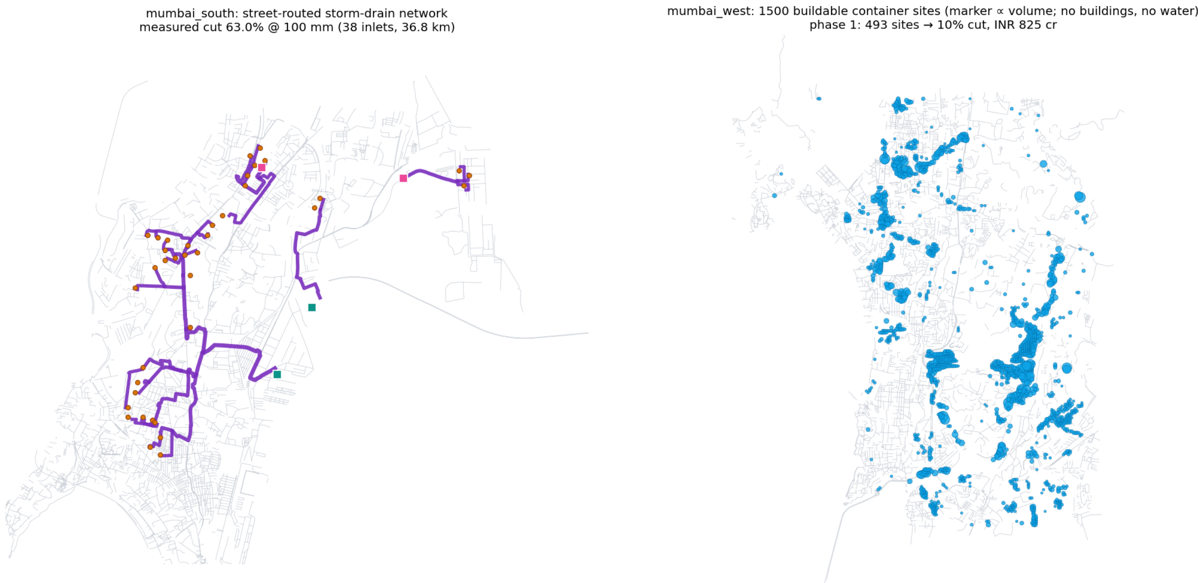

Figure 3: Mumbai tiles. **Left:** the street-routed storm-drain network for the Island City tile (purple, width $\propto$ accumulated flow; amber inlets, teal lowland / pink pit outfalls) concentrates on the Parel–Dadar–Byculla belt — the Hindmata neighbourhood where BMC actually built its underground detention tanks. **Right:** the 1,500 buildable container sites for the Western-suburbs tile (marker $\propto$ site volume) trace the Mithi corridor and the chronically waterlogged western drainage lines; phase 1 (493 sites) buys a measured 10% cut.

a per-node MLP) reaches only 0.866 on Patna versus 0.890 at 2 rounds and 0.906 at 4 — a monotone gain confirming that neighbourhood structure, not raw local topography, carries signal beyond what local features explain. This extends §3 to learned models: street-level flood knowledge is substantially city-transferable even where cell-level locations are not (and hardest into terrain regimes absent from training), because message passing aggregates exactly the local basin context that single cells lack.

Honesty and deployment. FloodGNN distils the twin, so it inherits the twin's biases: its claims are amortization, routing quality *under* the twin's flood fields, and transfer — not co-location skill, which §3 shows the twin itself largely lacks (and which a model learned directly from terrain does have, at 0.290 against the twin's 0.054). (Supervising directly on SAR-observed street water is the natural next step.) The $\sim$1.8 MB checkpoint ships inside the dashboard image; `/api/route` answers from the GNN where present and from the raster emulator otherwise, reporting its backend; a full-graph risk query takes 4 ms versus $\sim$260 ms for the raster-emulator pipeline it amortizes on the same hardware ($\sim$60$\times$).

6 A live participatory layer with an auditable learning gate

Everything above is computed from satellites and simulation; the people being flooded know things neither can see. Version 2 closes that loop under strict honesty constraints.

Citizen reports as data. Dashboard users drop a pin with a water level they can actually judge — ankle/knee/waist/chest, mapped to depth bands with explicit half-width tolerances ($\pm$0.15–0.35 m) — plus an optional note. Reports are validated against the area's model window, rate-limited by salted IP hash (5/hour, per-cell cool-down, global daily cap, a honeypot field for bots), held in memory for the live map, and mirrored by a background scheduler to a public dataset repository, which doubles as the durable store across free-tier restarts. No personal data is served: hashes never leave the backend.

Reports $\rightarrow$ supervision. A report is only a label together with the storm that caused it. The nightly job groups reports by IST calendar day, tags each day with the rainfall that actually fell (reanalysis archive, with the forecast API's recent-past window covering the archive's $\sim$5-day lag), drops dry days (<5 mm — a "knee-deep" report without rain is far more likely mischief or a burst pipe than a storm signal), and collapses

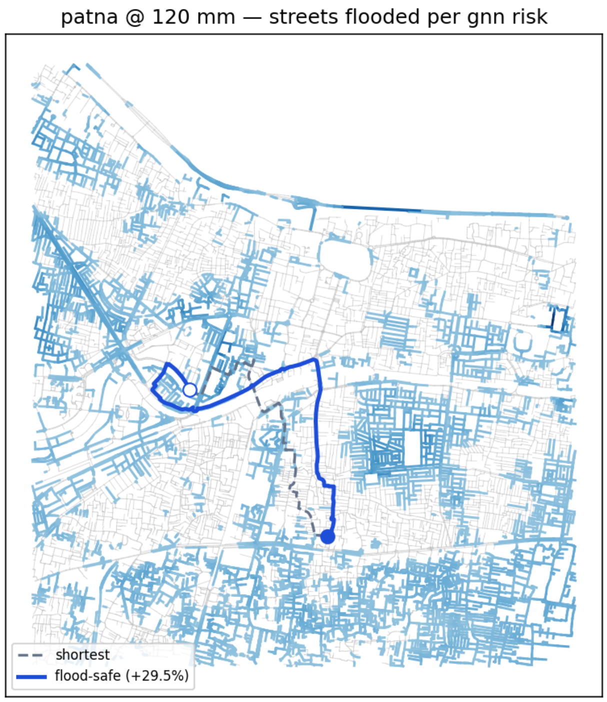

Figure 4: Flood-safe evacuation routing on the Patna street graph at 120 mm: streets shaded by GNN-predicted flood depth (blue = wet, grey = dry); the flood-ignorant shortest path (dashed) versus the FloodGNN route (solid). The learned route trades a modest detour for crossing a fraction of the floodwater — live on the public dashboard at any rainfall.

same-cell same-day reports to their median so one street corner cannot dominate.

The reward gate. Citizen reports carry no dry labels, so a model that predicts deep water everywhere would score perfectly on wet points — the same failure mode as the emulator's sparse targets, now in the field. A candidate emulator is fine-tuned with a band-tolerant Huber loss on report points (zero gradient inside the reported band) anchored by a simulation replay buffer in every batch, and it *ships only if all of*: $\geq$15 held-out points from $\geq$2 distinct storm-days (held out *by day* — points within a day are correlated); held-out point error improves $\geq$5%; replay RMSE degrades $\leq$10% (the all-wet guard); and it wins $\geq$70% of bootstrap resamples of the held-out errors. Every decision — accepted or held, with all four numbers — is appended to a public learning log surfaced on the dashboard, so the loop is auditable by its own users. At submission time the loop is deployed and gate-tested offline; no field update has yet cleared the gate (the reports dataset is days old), and per our own rules none should until the evidence bar is met.

Live context layers. The same discipline applies to the ambient layers: live-forecast alerts recompute the outlook read-only (never mutating committed artifacts); advisories are generated in English, Hindi and Marathi from *only* the system's own numbers (facts-JSON in, strict-JSON out, with a deterministic template fallback so the panel works without any LLM); and a VIIRS Black Marble [13] overlay marks built-up cells whose latest good-quality night radiance dropped >50% below their 90-day median. One trap is worth recording: during monsoon cloud cover, a fully quality-masked composite downloads as *zero radiance* — which naively reads as a city-wide blackout. Missing data must be sentinel-unmasked and treated as *unknown*, not dark.

7 Limitations and outlook

Hydraulics run at 60 m, coarser than the drawn street network; carved "drains" are full 60 m cells, so absolute excavation volumes are resolution artifacts even though per-metre drain costing is realistic. Depths are uncalibrated (and §3 shows why per-class physics calibration cannot fix co-location *within the solver*, though a learned model on the same rasters can); groundwater recharge site *rankings* use sample well data pending CGWB records, while the recharge *volumes* come from the twin's metered infiltration and do not depend on it. Mumbai adds regime-specific caveats: the tiles are independent closed-boundary models, so cross-seam flow is not simulated; tidal and storm-surge coupling — often the binding constraint for Mumbai outfalls — is absent, and rain falling on harbour cells ponds harmlessly inside the model rather than exchanging with the sea. SAR validation now covers Patna and both Mumbai tiles for which masks exist (29 storms in total); the DEM ensemble has been run for Patna and Bengaluru only. Mumbai-Harbour is excluded from headline validation because its observed water is almost entirely persistent — a storm increment of exactly zero on 8 of its 11 dates — so it measures a different hazard than urban pluvial flooding; a calendar-derived tide phase does not explain it (permutation $p = 0.47$), and a real tide gauge remains untested. Phase costs use a single indicative RCC rate; land, utilities and O&M are out of scope, so the ladders rank strategies rather than price projects. The learning loop's gate is deployed but unexercised by real reports; its first accepted (or correctly refused) field updates are the most interesting experiment we have not yet run. The single most valuable missing measurement is more cities: the transfer result of §3, on which the learned model's entire case rests, is computed over three domains. The natural next steps are therefore Sentinel-1 masks for more of the fourteen served areas, finer national DEMs (e.g. CartoDEM), assimilating mapped drain networks where they exist, sub-grid street conveyance, SAR-supervised FloodGNN (removing the "distils the twin" caveat), tidal boundaries for coastal tiles, and validating the *intervention deltas* (not just state) against before/after SAR where cities have actually built drainage.

Reproducibility

Code, all 16 per-area artifact bundles (including SAR masks, road graphs, replay buffers, and the ranked storage-site lists), 239 offline tests, and step-by-step runbooks: `https://github.com/Maverick-Ansh/project-varuna`. Live dashboard: `https://project-varuna-iota.vercel.app` (API: `anshvivek-varuna-floodtwin.hf.s` citizen-report dataset: `AnshVivek/varuna-reports` on Hugging Face. A new city builds with one script on a free Colab/Kaggle T4 and serves from the committed bundle; every headline number in this paper is regenerable from a committed artifact (the provenance table in `paper/README.md` maps each one to its file).

References

[1] Paul D. Bates, Matthew S. Horritt, and Timothy J. Fewtrell. A simple inertial formulation of the shallow water equations for efficient two-dimensional flood inundation modelling. *Journal of Hydrology*, 387(1–2): 33–45, 2010. doi: 10.1016/j.jhydrol.2010.03.027.

[2] Laurence Hawker, Peter Uhe, Luntadila Paulo, Jeison Sosa, James Savage, Christopher Sampson, and Jeffrey Neal. A 30 m global map of elevation with forests and buildings removed. *Environmental Research Letters*, 17(2):024016, 2022. doi: 10.1088/1748-9326/ac4d4f.

[3] Daniele Zanaga, Ruben Van De Kerchove, Wanda De Keersmaecker, et al. ESA WorldCover 10 m 2020 v100, 2021. European Space Agency.

[4] Jean-François Pekel, Andrew Cottam, Noel Gorelick, and Alan S. Belward. High-resolution mapping of global surface water and its long-term changes. *Nature*, 540:418–422, 2016. doi: 10.1038/nature20584.

[5] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241, 2015. doi: 10.1007/978-3-319-24574-4_28.

[6] Ramon Torres, Paul Snoeij, Dirk Geudtner, et al. GMES Sentinel-1 mission. *Remote Sensing of Environment*, 120:9–24, 2012. doi: 10.1016/j.rse.2011.05.028.

[7] Keith J. Beven and Michael J. Kirkby. A physically based, variable contributing area model of basin hydrology. *Hydrological Sciences Bulletin*, 24(1):43–69, 1979. doi: 10.1080/02626667909491834.

[8] Camilo D. Rennó, Antonio D. Nobre, Luz A. Cuartas, João V. Soares, Martin G. Hodnett, Javier Tomasella, and Maarten J. Waterloo. HAND, a new terrain descriptor using SRTM-DEM: Mapping terra-firme rainforest environments in Amazonia. *Remote Sensing of Environment*, 112(9):3469–3481, 2008. doi: 10.1016/j.rse.2008.03.018.

[9] OpenStreetMap contributors. OpenStreetMap, 2026. URL `https://www.openstreetmap.org`. Data available under the Open Database License.

[10] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *International Conference on Machine Learning (ICML)*, pages 1263–1272, 2017.

[11] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *AAAI Conference on Artificial Intelligence*, 2018.

[12] Thaine H. Assumpção, Ioana Popescu, Andreja Jonoski, and Dimitri P. Solomatine. Citizen observations contributing to flood modelling: opportunities and challenges. *Hydrology and Earth System Sciences*, 22 (2):1473–1489, 2018. doi: 10.5194/hess-22-1473-2018.

[13] Miguel O. Román, Zhuosen Wang, Qingsong Sun, Virginia Kalb, Steven D. Miller, Andrew Molthan, Lori Schultz, Jordan Bell, Eleanor C. Stokes, Bhartendu Pandey, Karen C. Seto, et al. NASA's Black Marble nighttime lights product suite. *Remote Sensing of Environment*, 210:113–143, 2018. doi: 10.1016/j.rse.2018.03.017.

[14] Patrick Zippenfenig. Open-Meteo.com weather API, 2023. Open-source weather API with free access for non-commercial use.